

FEDERICO BRUZZONE • CURRICULUM VITAE

PERSONAL INFORMATION

Born in Magenta (MI), Italy on 7th of **March 2000**
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CONTACT INFORMATION

github.com/FedericoBruzzone
[@federicobruzzone](https://twitter.com/federicobruzzone)
[in/federico-bruzzone](https://www.linkedin.com/in/federico-bruzzone)
[@fedebruzzone7](https://x.com/fedebruzzone7)
[u/FedericoBruzzone](https://www.youtube.com/u/FedericoBruzzone)

Presently a PhD Candidate in Computer Science. I have a strong passion for **Programming Languages** and **Compilers**, and I'm currently diving into **AI (ML/DL) compilers**. For more information, just visit my personal [webpage](#).

SCIENTIFIC PUBLICATIONS

- [1] F. Bruzzone, W. Cazzola, M. Brancaleoni, and D. Pellegrino, "Sink or SWIM: Tackling Real-Time ASR at Scale," *IEEE Transactions on Audio, Speech and Language Processing*, no. , pp. 1–11, 2026, doi: [10.1109/TASLPRO.2026.3675780](https://doi.org/10.1109/TASLPRO.2026.3675780). **Journal Ranked Q1 on Scimago** — [bib] [pdf] [IEEE] [arXiv]
[2] F. Bruzzone, W. Cazzola, and L. Favalli, "Code Less to Code More: Streamlining Language Server Protocol and Type System Development for Language Families," *Journal of Systems and Software*, p. 112554, Sept. 2025, doi: [10.1016/j.jss.2025.112554](https://doi.org/10.1016/j.jss.2025.112554). **Journal Ranked Q1 on Scimago** — [bib] [pdf] [SpringerLink] [arXiv]

PREPRINTS PUBLICATIONS

- [3] F. Bruzzone, W. Cazzola, and L. Favalli, "Generalized Software Product Line Extraction." [Online]. Available: <https://arxiv.org/abs/2605.28989> — [bib] [pdf] [arXiv]
[4] F. Bruzzone and W. Cazzola, "Meta-Monomorphizing Specializations." [Online]. Available: <https://arxiv.org/abs/2602.12973> — [bib] [pdf] [arXiv]
[5] F. Bruzzone, W. Cazzola, and L. Favalli, "From Separate Compilation to Sound Language Composition." [Online]. Available: <https://arxiv.org/abs/2602.03777> — [bib] [pdf] [arXiv]
[6] F. Bruzzone, W. Cazzola, M. Brancaleoni, and D. Pellegrino, "Sink or SWIM: Tackling Real-Time ASR at Scale." [Online]. Available: <https://arxiv.org/abs/2601.17097> — [bib] [pdf] [arXiv] **[PUBLISHED]**
[7] F. Bruzzone, W. Cazzola, and L. Favini, "Prioritizing Configuration Relevance via Compiler-Based Refined Feature Ranking." [Online]. Available: <https://arxiv.org/abs/2601.16008> — [bib] [pdf] [arXiv]
[8] F. Bruzzone, W. Cazzola, and L. Favalli, "Code Less to Code More: Streamlining Language Server Protocol and Type System Development for Language Families," *Journal of Systems and Software*, p. 112554, Sept. 2025, doi: [10.1016/j.jss.2025.112554](https://doi.org/10.1016/j.jss.2025.112554). — [bib] [pdf] [arXiv] **[PUBLISHED]**

EDUCATION

2024-Present	PhD Candidate in Computer Science at the ADAPT Lab , University of Milan
2022-2024	MSc in Computer Science at University of Milan (<i>110/110 cum laude</i> , W. Cazzola, 15/07/2024) Thesis: " Toward a Modular Approach for Type Systems and LSP Generation ".
2019-2022	BSc in Musical Computer Science at University of Milan (13/10/2022)
2011-2019	Piano and Music Composition at I.S.S.M. Novara Conservatory
2014-2019	Diploma in Computer Science and Telecommunications at E. Alessandrini

OPEN SOURCE CONTRIBUTIONS (SELECTED)

2026	The Eter Programming Language , a new programming language leveraging the MLIR/IREE ecosystem with tensors support.
2026	MLIR/LLVM: [mlir][vector] Add opt-in inbounds/nneg flags to vector.load/vector.store
2026	MLIR/LLVM: [mlir][linalg] add toContractionDimensions for healthy code reuse
2026	MLIR/LLVM: [mlir][vector] extend createReadOrMaskedRead/createWriteOrMaskedWrite with permutation map support
2026	MLIR/LLVM: [mlir][affine] emit in_bounds on transfer_read/write when statically provable in affine-super-vectorize
From 2026	Maintainer of the LLVM Pass Template , a C++ template project to quickly create new LLVM passes.
From 2025	Maintainer of the Papers on Compiler Optimizations: Analysis and Transformations , a curated list of scientific publications
From 2025	Maintainer of scribe , a minimalist, opinionated LaTeX document class for technical writing and presentations.

From 2025 Maintainer of the [Tide Compiler](#), an backend-agnostic IR and compiler framework written in **Rust**.
2025 [Generalizing Closeness centrality to weighted networks using Newman method](#) (issue 1384)
From 2024 Maintainer of the cross-platform [tgt](#) project and [tdlib-rs](#) TDLib bindings written in **Rust**.

RESEARCH ACTIVITIES

2024-Present **MUSEMI Session Chair, Speaker and Co-organizer**
May 2026 **Committee Member** at the Int. Conf. on Software Language Engineering ([SLE 2026](#)), **ACM, B on CORE**
Nov 2025 **Reviewer** for the Journal of Computer Languages ([COLA](#)), **Elsevier, C on CORE**
Jun 2025 **Reviewer** for the Journal of Software and Systems Modeling ([SoSyM](#)), **Springer, Q1 on Scimago**
Dec 2025 **Reviewer** for the Eur. Conf. on Object Oriented Programming ([ECOOP 2026](#)), **ACM, A on CORE**
2-6 Jun 2025 **Participant** at [<Programming> 2025](#) conference
Apr-Jun 2025 **Committee Member** at the Int. Conf. on Software Language Engineering ([SLE 2025](#)), **ACM, B on CORE**
Mar 2025 **Reviewer** for the Journal of Systems and Software ([JSS](#)), **Elsevier, Q1 on Scimago**
2-7 Sep 2024 **Student Volunteer** at the Int. Conf. on Functional Programming ([ICFP 2024](#))
2022-2024 Research **Internship** at [ADAPT Lab](#), working on modular type systems and LSP generation for [Neverlang](#)
2021-2022 Research **Internship** at [LIM Lab](#), working on the [IEEE 1599](#) standard

TEACHING ACTIVITIES

THESIS SUPERVISION

23/02/2026 D. Cerato, *Employing Metaprogramming: A Macro-Driven Strategy to Overcome Compiler-Imposed Limitations on Specialization*, BSc, **103**
10/04/2025 D. Pellegrino, *Scalable Multi-client Real-time Whisper*, BSc, **96**
09/04/2025 A. Longoni, *GUIDE: Graphical User Interface Development Environment*, MSc, **110L**
09/04/2025 L. Albani, *New Generalized Protocol For Software Product Line Extraction And Configuration*, MSc, **110L**
09/04/2025 G. Esposito, *Fr3D: A Framework for DAP-compatible DSL-oriented Debugging*, MSc, **110L**
24/02/2025 L. Favini, *RustyEx: Intrumenting rustc to Extract Feature Dependency Graphs*, BSc, **102**

GRADUATE COURSES

2025-2026 Mathematical Logic (Art. 45), **BSc** in CS, University of Milan, *S. Aguzzoli*
2024-2025 Programming 1 (Art. 45), **BSc** in CS, University of Milan, *L. Capra* (coordinator W. Cazzola)
2024-2025 Mathematical Logic (Art. 45), **BSc** in CS, University of Milan, *S. Aguzzoli*
2023-2024 Mathematical Logic, **BSc** in CS, University of Milan, *S. Aguzzoli*
2023-2024 Computer Science, **BSc** in Comm. and Society, University of Milan, *A. Momigliano*
2023-2024 Programmin 1, **BSc** in CS, University of Milan, *A. Trentini* (coordinator P. Boldi)
2023-2024 Programming in Python, **MSc** in Chemistry, University of Milan, *M. Monga*

ADDITIONAL ACTIVITIES

2023-Present **Private Tutoring** in Computer Science, Mathematics and Physics
Nov 2024 **Collaborator** at the [Bebras Challenge](#) , [ALaDDIn Lab](#)
Jan-Jun 2024 **Organizer** of the BSc Computer Science Laboratories, [ALaDDIn Lab](#)
Jan-Jun 2024 **Collaborator** of the Workshops for schools, [ALaDDIn Lab](#)
Nov 2023 **Collaborator** at the [Bebras Challenge](#) , [ALaDDIn Lab](#)

TECHNICAL SKILLS

Languages Rust, C/C++, Python, Go, OCaml, Java, Scala, Kotlin, Erlang, Lua, Dart, PHP, HTML/CSS, SQL, Bash, TeX
Systems/Tooling Git, CI/CD, Docker, GDB, Valgrind, Build Systems (e.g., CMake, Cargo, Pip), Cross-language Linking, Static/Dynamic Libraries, and FFI (e.g., C bindgen)
Area of Expertise Compiler Construction and Optimizations, Programming Languages Design, IR Design and Implementation, Type Systems, Language Support Tools (e.g., LSP), Static Analysis, Rustc Internals, MLIR, LLVM, Parsing Techniques and Parser Generators (e.g., ANTLR)

MUSICAL ACTIVITIES

2021-Present **Sound Engineer and Music Producer**, in collaboration with [Believe](#) and as a [SIAE](#) member.
2006-Present **Pianist and Music Composer**, composing original pieces and arrangements for piano and various ensembles
2019-Present **Piano and Music Teacher**, teaching piano and music theory and composition to students of all ages and levels

GRANTS AND FELLOWSHIPS

2024	56th Top Github Public Contributor in Italy out of 958
2023-2024	Scholarship for the MSc in Computer Science, awarded by the University of Milan
2020-2024	Scholarship for the BSc in Musical Computer Science, awarded by the University of Milan

TONGUES

Italian	Mother tongue
English	Level CEFR B2 (SLAM at University of Milan)
Spanish	Base (A1-A2)

Milan, 15/06/2026